

# Le système international d'unités The International System of Units

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Annexe 2

## Annex 2

Valeurs recommandées des fréquences étalons  
destinées à la mise en pratique de la définition du  
mètre et aux représentations secondaires de la  
seconde

**Recommended values of standard frequencies  
for applications including the practical  
realization of the metre and secondary  
representations of the second**

Partie 2

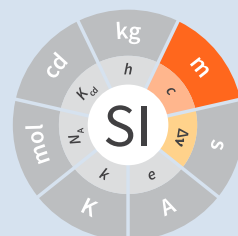
## Part 2

Rubidium 87 Atom ()

**Rubidium 87 Atom ()**

CCL-CCTF Frequency Standards Working Group

May 2016  
mai 2016





**Recommended values of standard  
frequencies for applications including  
the practical realization of the metre and  
secondary representations of the second  
Part 2: Rubidium 87 Atom ()**

CCL-CCTF Frequency Standards Working Group

**9<sup>th</sup> edition      2019**

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## 1. Recommended value [1] of the frequency:

$$f(^{87}\text{Rb}) = 6\,834\,682.610\,904\,312\,6\text{ Hz}$$

with a relative standard uncertainty of  $6 \times 10^{-16}$ .

This radiation was endorsed by the CIPM as a secondary representation of the definition of the second [2].

## 2. Source data

The adopted value of  $f(^{87}\text{Rb}) = 6\,834\,682.610\,904\,312\,6\text{ Hz}$  with a relative standard uncertainty of  $6 \times 10^{-16}$  has been calculated by the methods presented in [3], [4], [5], [6], with the data from the table of frequencies (2017) attached below. Given that there are only two contributors and the result is heavily influenced by the LNE-SYRTE Rb standard by both direct frequency and frequency ratio measurements, which have a total uncertainty of  $3.1 \times 10^{-16}$ , the CCTF multiplied the total uncertainty by a factor of 2 to give  $6.2 \times 10^{-16}$  and rounded it to  $6 \times 10^{-16}$  following the procedures outlined in ref. [6].

**Table 1.** Table of evaluated frequencies (2017)

| Frequency / Hz           | Uncertainty / Hz | Reference  |
|--------------------------|------------------|------------|
| $^{199}\text{Hg}$        |                  |            |
| 1 128 575 290 808 155.1  | 6.4              | [7]        |
| 1 128 575 290 808 154.62 | 0.41             | [8]        |
| $^{171}\text{Yb}$        |                  |            |
| 518 295 836 590 864.0    | 28.0             | [38]       |
| 518 295 836 590 863.1    | 2.0              | [39]       |
| 518 295 836 590 865.2    | 0.7              | [40]       |
| 518 295 836 590 863.5    | 8.1              | [41]       |
| 518 295 836 590 863.59   | 0.31             | [42]       |
| 518 295 836 590 863.38   | 0.57             | [43]       |
| $^{88}\text{Sr}$         |                  |            |
| 429 228 066 418 009.0    | 32.0             | [9]        |
| 429 228 066 418 008.3    | 2.1              | [10], Sr1  |
| 429 228 066 418 007.3    | 2.9              | [10], Sr2  |
| $^{87}\text{Sr}$         |                  |            |
| 429 228 004 229 874.0    | 1.1              | [11]       |
| 429 228 004 229 873.65   | 0.37             | [12]       |
| 429 228 004 229 873.6    | 1.1              | [13]       |
| 429 228 004 229 874.1    | 2.4              | [14]       |
| 429 228 004 229 872.9    | 0.5              | [15]       |
| 429 228 004 229 873.9    | 1.4              | [16]       |
| 429 228 004 229 872.0    | 1.6              | [17]       |
| 429 228 004 229 873.56   | 0.49             | [18]       |
| 429 228 004 229 873.7    | 1.4              | [19]       |
| 429 228 004 229 873.13   | 0.17             | [20]       |
| 429 228 004 229 873.10   | 0.13             | [21]       |
| 429 228 004 229 872.92   | 0.12             | [22]       |
| 429 228 004 229 872.97   | 0.16             | [23], 2014 |

| Frequency / Hz          | Uncertainty / Hz     | Reference       |
|-------------------------|----------------------|-----------------|
| 429 228 004 229 873.04  | 0.11                 | [23], June 2015 |
| 429 228 004 229 872.97  | 0.40                 | [24]            |
| 429 228 004 229 872.99  | 0.18                 | [25]            |
| $^{40}\text{Ca}^+$      |                      |                 |
| 411 042 129 776 393.2   | 3.0 <sup>(a)</sup>   | [26]            |
| 411 042 129 776 398.4   | 1.2                  | [27]            |
| 411 042 129 776 400.5   | 1.2                  | [28], 2012      |
| 411 042 129 776 401.7   | 1.1                  | [28], 2014/2015 |
| $^{87}\text{Rb}$        |                      |                 |
| 6 834 682 610.904 312 9 | $2.4 \times 10^{-6}$ | [29]            |
| 6 834 682 610.904 307 0 | $3.1 \times 10^{-6}$ | [30]            |
| 6 834 682 610.904 312 5 | $2.1 \times 10^{-6}$ | [31]            |

(a) From the least square procedure it turned out that the value and the uncertainty given in ref. [26] is not compatible with the remaining data. Thus the uncertainty given in the original publication [26] was increased to 3 Hz to make it statistically more consistent.

**Table. Table of evaluated frequencies (2017) (cont.)**

| Frequency ratios                  | Fractional uncertainty  | Reference  |
|-----------------------------------|-------------------------|------------|
| $^{199}\text{Hg}/^{87}\text{Sr}$  |                         |            |
| 2.629 314 209 898 909 60          | $8.4 \times 10^{-17}$   | [32]       |
| 2.629 314 209 898 909 15          | $1.8 \times 10^{-16}$   | [8]        |
| $^{199}\text{Hg}/^{87}\text{Rb}$  |                         |            |
| 165 124.754 879 997 258           | $3.8 \times 10^{-16}$   | [8]        |
| $^{171}\text{Yb}/^{87}\text{Sr}$  |                         |            |
| 1.207 507 039 343 341 2           | $1.4 \times 10^{-15}$   | [33], [34] |
| 1.207 507 039 343 337 76          | $2.4 \times 10^{-16}$   | [35]       |
| 1.207 507 039 343 337 749         | $4.6 \times 10^{-17}$   | [36]       |
| $^{88}\text{Sr}/^{87}\text{Sr}$   |                         |            |
| 1.000 000 144 883 682 777         | $7 \times 10^{-17}$ (a) | [37]       |
| $^{87}\text{Sr}/^{40}\text{Ca}^+$ |                         |            |
| 1.044 243 334 529 641 6           | $2.3 \times 10^{-15}$   | [27]       |
| $^{87}\text{Sr}/^{87}\text{Rb}$   |                         |            |
| 62 801.453 800 512 435            | $3.3 \times 10^{-16}$   | [22]       |

(a) The fractional uncertainty of  $2.3 \times 10^{-17}$  in ref. [37] has been increased by a factor of three since this uncertainty is an order of magnitude smaller than the other input data for this transition.

**$^{87}\text{Rb}$ , unperturbed ground-state hyperfine transition**

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